

CSE563 — Concept Sheet, Block B

Unit III: Probability and Statistics

Theory reference — every concept, no numericals

School of Computer Science & Engineering

Session 2025–26



What This Sheet Is

A **theory-only** reference for Unit III, taught as a standalone block because probability and statistics share no machinery with the units either side of them. Every definition, law, theorem and interpretive statement that can be examined in written form appears here. Calculations live in the *Unit III Numerical Guide*; practice questions live in the *Unit III Worksheet*.

Contents

1 Basic Probability

Foundational Definitions

Random experiment — a process whose outcome cannot be predicted with certainty.

Sample space S — the set of all possible outcomes.

Event — any subset $E \subseteq S$. An *elementary* event is a single outcome.

Mutually exclusive events — $A \cap B = \emptyset$ — they cannot occur together.

Exhaustive events — their union is the whole sample space.

Partition — a collection of mutually exclusive, exhaustive events — the setting required for the law of total probability and for Bayes' theorem.

Kolmogorov's Axioms and Their Consequences

Axioms. (1) $0 \leq P(E) \leq 1$ for every event. (2) $P(S) = 1$. (3) For mutually exclusive $E_1, E_2, \dots$, $P(\bigcup E_i) = \sum P(E_i)$.

Derived results.

- $P(\bar{E}) = 1 - P(E)$ — the *complement rule*, which is the fastest route to any “at least one” question.
- $P(\emptyset) = 0$.
- Monotonicity: $A \subseteq B \Rightarrow P(A) \leq P(B)$.
- **Addition rule:** $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. This is inclusion–exclusion (Unit I) transferred to probability — events *are* sets.

Three Interpretations of Probability

Classical — $P(E) = |E|/|S|$, valid only when outcomes are equally likely. Requires counting techniques: the product rule, permutations $P(n, r) = \frac{n!}{(n-r)!}$ and combinations $\binom{n}{r}$.

Empirical (frequentist) — the long-run relative frequency, $P(E) = \lim_{n \rightarrow \infty} (\text{occurrences})/n$. The basis of the frequentist school.

Subjective (Bayesian) — a degree of belief, revisable in the light of evidence. The basis of the Bayesian school.

2 Compound Probability

Conditional Probability and Independence

Conditional probability — $P(A | B) = \frac{P(A \cap B)}{P(B)}$ for $P(B) > 0$. Conditioning *shrinks the sample space* to B .

Multiplication rule — $P(A \cap B) = P(A | B)P(B) = P(B | A)P(A)$.

Chain rule — $P(A_1 \cap A_2 \cap A_3) = P(A_1)P(A_2 | A_1)P(A_3 | A_1 \cap A_2)$.

Independence — A and B are independent iff $P(A \cap B) = P(A)P(B)$, equivalently $P(A | B) = P(A)$ — one event carries no information about the other.

Law of total probability — for a partition $B_1, \dots, B_k$: $P(A) = \sum_i P(A | B_i)P(B_i)$.

Mutually Exclusive Is Not Independent

This is the single most-confused pair in the unit.

	Mutually exclusive	Independent
Condition	$P(A \cap B) = 0$	$P(A \cap B) = P(A)P(B)$
Meaning	Cannot both occur	Neither informs the other
$P(A B)$	0	$P(A)$

Two events of non-zero probability that are mutually exclusive are *maximally dependent*: learning that B occurred tells you with certainty that A did not.

Bayes' Theorem

$$P(B_j | A) = \frac{P(A | B_j)P(B_j)}{P(A)} = \frac{P(A | B_j)P(B_j)}{\sum_i P(A | B_i)P(B_i)}.$$

What it does — **reverses the direction of conditioning** — from “probability of the evidence given the cause” to “probability of the cause given the evidence”.

Prior $P(B_j)$ — belief before seeing evidence. **Likelihood** $P(A | B_j)$: how well the cause explains the evidence. **Posterior** $P(B_j | A)$: the updated belief.

Base-rate fallacy — the classic error of ignoring the prior. A test that is “99% accurate” for a disease affecting 1% of people still yields mostly false positives, because the healthy population is 99 times larger. The same arithmetic governs spam filters, intrusion detection and fraud alerts — any rare-event detector.

Check — the posteriors over a partition must sum to 1.

3 Random Variables and Probability Distributions

Random Variables

Random variable — a function from the sample space to $\mathbb{R}$.

Discrete — takes countably many values; described by a *probability mass function* $p(x) = P(X = x)$ with $\sum p(x) = 1$.

Continuous — takes values in an interval; described by a *probability density function* $f(x)$ with $P(a \leq X \leq b) = \int_a^b f$. For a continuous variable $P(X = x) = 0$ for any single x .

Cumulative distribution function — $F(x) = P(X \leq x)$, non-decreasing from 0 to 1.

Expectation and Variance

$$E[X] = \sum_x xp(x) \text{ or } \int xf(x) dx, \quad \text{Var}(X) = E[(X - \mu)^2] = E[X^2] - \mu^2.$$

Linearity of expectation — $E[aX + b] = aE[X] + b$, and $E[X + Y] = E[X] + E[Y]$ *always* — even for dependent variables.

Variance rules — $\text{Var}(aX + b) = a^2 \text{Var}(X)$ (adding a constant shifts but does not spread). $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ *only if independent*.

Standard deviation — $\sigma = \sqrt{\text{Var}(X)}$, carrying the same units as X — which is why it, rather than the variance, is quoted in reports.

Fair game — one with $E[\text{gain}] = 0$.

The Standard Distributions

Distribution	Models	Mean	Variance	Recognition cue
Bernoulli(p)	one yes/no trial	p	$p(1-p)$	a single success/failure
Binomial(n, p)	successes in n independent trials	np	$np(1-p)$	<i>fixed</i> n , constant p
Geometric(p)	trials until the first success	$1/p$	$(1-p)/p^2$	“how many attempts until”
Poisson(λ)	counts of rare events per interval	λ	λ	a <i>rate</i> , no fixed n
Uniform(a, b)	equal likelihood on an interval	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	“no information” between limits
Exponential(λ)	waiting time between Poisson events	$1/\lambda$	$1/\lambda^2$	time <i>until</i> an event; memoryless
Normal(μ, σ^2)	sums/averages of many effects	μ	σ^2	a bell-shaped continuous measurement

Approximations — Binomial with large n and small $p \approx \text{Poisson}(np)$. Binomial with large $n \approx \text{Normal}(np, np(1-p))$.

Scaling Poisson — λ must be rescaled in proportion to the interval considered — the commonest error in the unit.

The Normal Distribution

Standardisation — if $X \sim N(\mu, \sigma^2)$ then $Z = \frac{X - \mu}{\sigma} \sim N(0, 1)$, allowing a single table to serve every normal distribution.

Symmetry — $\Phi(-z) = 1 - \Phi(z)$.

Empirical rule — about 68% of the distribution lies within $\pm 1\sigma$, 95% within $\pm 2\sigma$, 99.7% within $\pm 3\sigma$.

Percentiles — to find the value exceeded by α of the population, look up z with $\Phi(z) = 1 - \alpha$ and compute $x = \mu + z\sigma$. In systems engineering this is how “p90” and “p99” latency figures are defined.

4 Frequentist Inference

Population, Sample, Estimator

Parameter — a fixed but unknown numerical property of the population (μ, σ, p).

Statistic — a quantity computed from the sample ($\bar{x}, s, \hat{p}$).

Estimator — a statistic used to estimate a parameter. It is *unbiased* if its expected value equals the parameter — which is why the sample variance divides by $n - 1$, not n .

The frequentist stance — the parameter is a fixed constant; all randomness lies in the sampling. Every probability statement therefore describes *what would happen over repeated sampling*.

Central Limit Theorem and Standard Error

CLT. For independent, identically distributed samples of size n from *any* distribution with mean μ and finite variance σ^2 ,

$$\bar{X} \approx N\left(\mu, \frac{\sigma^2}{n}\right) \quad \text{for large } n \text{ (rule of thumb } n \geq 30\text{)}.$$

Why it matters — it licenses normal-based inference regardless of the shape of the underlying population — the reason the normal distribution dominates statistics.

Standard error — $SE = \sigma/\sqrt{n}$, the standard deviation of the sampling distribution of $\bar{X}$. It measures how much $\bar{x}$ would vary between repeated samples — not how much the data vary.
The $\sqrt{n}$ law — to halve the uncertainty you must *quadruple* the sample. This governs the economics of every survey and experiment.

Confidence Intervals

$$\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \quad (\sigma \text{ known or } n \text{ large}), \quad \bar{x} \pm t_{\alpha/2, n-1} \frac{s}{\sqrt{n}} \quad (\sigma \text{ unknown, small } n).$$

Critical values: 90% $\rightarrow$ 1.645, 95% $\rightarrow$ 1.96, 99% $\rightarrow$ 2.58.

Correct interpretation — the *procedure* captures the true parameter in $100(1 - \alpha)\%$ of repeated samples.

Incorrect interpretation — “there is a 95% probability that μ lies in this interval”. In the frequentist framework μ is a constant — it is either in this particular interval or it is not. A direct probability statement about the parameter requires a Bayesian *credible* interval.

Trade-off — higher confidence widens the interval; larger n narrows it.

Hypothesis Testing — the Five Steps

1. State H_0 (the “no effect” claim, always an equality) and H_1 (two-sided $\neq$, or one-sided $<$ / $>$ if the wording is directional).
2. Choose the significance level α .
3. Compute the test statistic — z if σ is known or n is large, t on $n - 1$ degrees of freedom otherwise.
4. Compare with the critical value, or compute the p -value.
5. Decide and **state the conclusion in context**.

p -value — the probability, *assuming H_0 is true*, of observing a statistic at least as extreme as the one obtained. Small p means the data are surprising under H_0 .

Never say “accept H_0 ” — failing to reject means the evidence was insufficient, not that H_0 is proven. Absence of evidence is not evidence of absence.

Duality with intervals — a two-sided test at level α rejects exactly those null values lying outside the $100(1 - \alpha)\%$ confidence interval.

Errors and Power

	H_0 is true	H_0 is false
Reject H_0	Type I error , probability α (false alarm)	Correct decision, probability $1 - \beta$ (<i>power</i>)
Fail to reject	Correct decision, probability $1 - \alpha$	Type II error , probability β (a miss)

The trade-off — at fixed n , lowering α raises β . Only increasing the sample size reduces both.

Choosing which error to fear — context decides. In cancer screening a missed case (Type II) is far worse than a false alarm that a follow-up test can clear, so α is deliberately set high. In a criminal trial, convicting the innocent (Type I) is the graver error.

Multiple testing — testing many hypotheses inflates the false-positive rate; the Bonferroni correction uses α/m for m tests.

Other Tests Worth Naming

Two-sample t -test — compares the means of two independent groups — the standard A/B test.

Paired t -test — same subjects measured twice (before/after an optimisation).

Chi-square test of independence — two categorical variables; statistic $\chi^2 = \sum(O - E)^2/E$.
ANOVA — compares the means of three or more groups simultaneously.

5 Bayesian Inference

The Bayesian Recipe

Treat the parameter θ as a *random variable* carrying a distribution:

$$\underbrace{P(\theta | D)}_{\text{posterior}} \propto \underbrace{P(D | \theta)}_{\text{likelihood}} \times \underbrace{P(\theta)}_{\text{prior}}.$$

Output — a full distribution over θ , not a single number. Summarised by the posterior mean, the posterior mode (**MAP**), or a **credible interval**.

Credible interval — unlike a confidence interval, a 95% credible interval *does* mean “there is a 95% probability θ lies here”.

Sequential updating — today’s posterior is tomorrow’s prior, making Bayesian inference naturally online — the basis of Kalman filters and adaptive spam filters.

Conjugacy and Point Estimates

Conjugate prior — a prior whose family is preserved by the update, giving a closed-form posterior. The key example is **Beta–Binomial**: a $\text{Beta}(\alpha, \beta)$ prior with k successes in n trials yields a $\text{Beta}(\alpha + k, \beta + n - k)$ posterior. Others: Gamma–Poisson, Normal–Normal.

Pseudo-counts — α and β behave like prior successes and failures, which is how the strength of a prior is judged against the sample size.

MLE — $\arg \max_{\theta} P(D | \theta)$ — uses the data only.

MAP — $\arg \max_{\theta} P(D | \theta)P(\theta)$ — weights the likelihood by the prior. **MAP = MLE when the prior is uniform**, and the two converge as n grows.

Shrinkage — the posterior is pulled from the data estimate toward the prior mean, by an amount decreasing with sample size. This is a feature: it stabilises estimates from small samples.

Frequentist versus Bayesian — the Comparison

Question	Frequentist	Bayesian
What is θ ?	A fixed unknown constant	A random variable with a distribution
Where is the randomness?	In the data (repeated sampling)	In our uncertainty about θ
Point estimate	MLE	MAP or posterior mean
Interval	Confidence (long-run coverage)	Credible (direct probability)
Prior knowledge	Not used	Explicit, via the prior
Small samples	Wide, unstable intervals	Regularised by the prior
Computational cost	Usually closed-form	May require sampling (MCMC)

Where they meet. MAP estimation with a Gaussian prior is mathematically identical to L2-regularised (ridge) regression — the two schools reach the same formula by different reasoning.

Naive Bayes — the Classifier

$$P(C | \mathbf{x}) \propto P(C) \prod_{i=1}^p P(x_i | C).$$

The “naive” assumption — features are conditionally independent given the class. This is

almost always false in reality — yet the classifier performs well, because accurate *ranking* of classes does not require accurate probabilities.

Laplace (add-one) smoothing — prevents a single unseen feature from zeroing the entire product. It is a Beta(1, 1)-style prior in disguise.

Uses — spam filtering, document classification, sentiment analysis — anywhere features are numerous and data sparse.

6 Regression Analysis

Correlation

Pearson correlation coefficient — $r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \in [-1, 1]$, measuring the strength and direction of a *linear* association.

Interpretation — $r = \pm 1$ perfect line; $r = 0$ no *linear* relation — a strong non-linear relationship can still give $r \approx 0$.

Correlation is not causation — ice-cream sales and drowning correlate through a *confounder* (temperature). Causal claims require a randomised experiment, not a correlation coefficient.

Simple Linear Regression

Model: $y = \beta_0 + \beta_1 x + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$. Least squares minimises $\sum (y_i - \hat{y}_i)^2$, giving

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}, \quad \hat{\beta}_1 = r \frac{s_y}{s_x}.$$

Two free checks — the fitted line always passes through $(\bar{x}, \bar{y})$, and the residuals always sum to zero.

Interpreting the slope — a one-unit increase in x is *associated with* a $\hat{\beta}_1$ change in y — never “causes”, unless the data came from a designed experiment.

Interpreting the intercept — the fitted value at $x = 0$, which is often meaningless if $x = 0$ lies far outside the observed range.

Goodness of Fit and Inference

$$\text{SST} = \sum (y_i - \bar{y})^2, \quad \text{SSE} = \sum (y_i - \hat{y}_i)^2, \quad R^2 = 1 - \frac{\text{SSE}}{\text{SST}}.$$

Meaning of R^2 — the proportion of variation in y explained by the model. For *simple* regression $R^2 = r^2$.

Adjusted R^2 — $1 - (1 - R^2) \frac{n-1}{n-p-1}$. Necessary because plain R^2 *always* rises when a predictor is added, however useless.

Testing the slope — $t = \hat{\beta}_1 / \text{SE}(\hat{\beta}_1)$ on $n - 2$ degrees of freedom tests $H_0 : \beta_1 = 0$ — “is there any linear relationship at all?”

Multiple regression — $\hat{\beta} = (X^T X)^{-1} X^T \mathbf{y}$ — Unit II linear algebra applied directly to Unit III statistics. $X^T X$ is singular under perfect collinearity.

Assumptions and Abuses

The LINE assumptions: Linearity, Independence of errors, Normality of residuals, Equal variance (homoscedasticity).

Residual plot — predicted values against residuals diagnoses two at once — a visible curve means linearity is violated; a funnel shape means heteroscedasticity.

Extrapolation — predicting outside the observed range of x is unsupported by the data and should always be flagged.

Outliers and leverage — a point with an extreme x value exerts disproportionate influence on the fitted line.

Silent failure — software will fit a linear model to data violating every assumption and issue no warning whatsoever. The diagnostics are the analyst's responsibility.

Logistic Regression — Regression for Classification

When $y \in \{0, 1\}$, model the *probability* rather than the value:

$$P(y = 1 \mid \mathbf{x}) = \sigma(\boldsymbol{\beta}^T \mathbf{x}) = \frac{1}{1 + e^{-\boldsymbol{\beta}^T \mathbf{x}}}, \quad \text{equivalently} \quad \ln \frac{p}{1-p} = \boldsymbol{\beta}^T \mathbf{x}.$$

Name trap — despite “regression” in its name, logistic regression is a **classification** algorithm. Examiners test this.

Fitting — by maximum likelihood; there is no closed form, so gradient descent is used — the Unit I calculus, with $\sigma'(x) = \sigma(x)(1 - \sigma(x))$.

Coefficient meaning — a one-unit increase in x_j multiplies the *odds* by e^{β_j} .

💡 Exam Tip & Key takeaway: Unit III — The Threads Worth Seeing

- **Events are sets.** The addition rule *is* inclusion–exclusion from Unit I; independence is a statement about intersections.
- **Bayes is one line of algebra** rearranged from the multiplication rule — but its consequences (base rates, spam filters, medical testing) are the most practically important content in the unit.
- **The CLT is why the normal distribution is everywhere**, and $SE = \sigma/\sqrt{n}$ is why data collection obeys diminishing returns.
- **Confidence intervals and two-sided tests are the same statement** viewed from two directions.
- **Multiple regression is linear algebra:** $(X^T X)^{-1} X^T \mathbf{y}$ is Unit II solving an overdetermined system in the least-squares sense.
- **Frequentist and Bayesian answers converge** as the sample grows; they differ most, and matter most, when data are scarce.